

Distance between two points



1 a $AB = 3$ units b $AB = 5$ units c $AB = 10$ units d $AB = 5$ units

2 a $AB = 6$ units b $AB = 3$ units c $AB = 14$ units d $AB = 8$ units

3 a $AC = 5$ units b $AC = 3.61$ units

4 a i $PQ = 3$ units ii $QR = 4$ units iii $PR = 5$
b i $PQ = 3$ units ii $QR = 4$ units iii $PR = 5$
c i $PQ = 4$ units ii $QR = 7$ units iii $PR = 8.06$
d i $PQ = 5$ units ii $QR = 4$ units iii $PR = 6.40$

5 a i $AB = 5$ units ii $BC = 12$ units
 iii $c = 13$
b i $AB = 11$ units ii $BC = 7$ units
 iii $c = 13.0$

6 11.31 units

Distance formula

7 a 20 b 20 c 13.04 d 6.40

8 a $AC = 5.39$ b $AC = 5$ c $AC = 6.40$ d $AC = 7.81$
e $AC = 11.40$ f $AC = 17$



- 9 a 10 units b 15 units c $\sqrt{73}$ units d $\sqrt{34}$ units
- 10 a $\sqrt{82}$ units b $\sqrt{26}$ units c N
- 11 a $AB = 9.90$ units b $PN = 11.18$ units
c N
- 12 a $AB = 3.61$ units b $AM = 1.80$ units
- 13 $M(-2, 9)$ and $M(-2, 3)$
- 14 a $AC = \sqrt{13}$ b $BC = \sqrt{13}$
c C is equidistant from A and B .
- 15 $AK = 7.2$ units

Geometric applications

- 16 a $BC = 6$ b $AB = \sqrt{18}$ c $AC = \sqrt{18}$ d Isosceles
- 17 a i $AC = \sqrt{200}$ units ii $BD = \sqrt{72}$ units
b 60 units^2
- 18 a 17 units
b 34π units
c 17 units
d Yes, because the distance between the centre and the point $(20, 4)$ is the same as the radius.

19 a

i $AB = 5$ units

iii $BC = 8$ units

b Isosceles



ii $AC = 5$ units

20

a

- Check if the sides satisfy Pythagoras' Theorem.
- Check if two sides are equal so that the triangle must be isosceles.

b

i $AB = \sqrt{85}$ units

ii $AC = \sqrt{85}$ units

iii $BC = \sqrt{170}$ units

c Yes

21

a

i $PQ = \sqrt{242}$ units

ii $QR = \sqrt{340}$ units

iii $PR = \sqrt{98}$ units

b Yes

22

a

i $MN = \sqrt{8}$ units

ii $NR = \sqrt{50}$ units

b 2 : 5

23 23.1 units

24

a 50 units^2

b $\sqrt{200}$ units

c $d = 5\sqrt{2}$

25

a $10x$

b $(7x, -2x)$

26

a $3\sqrt{10}$

b $4\sqrt{2}$

c $3\sqrt{10}$

d $4\sqrt{2}$

e $\sqrt{218}$

f $\sqrt{26}$

g Parallelogram